

CNC Alpha-1 — Operation & Maintenance Manual

Overview

The CNC Alpha-1 is a 3-axis vertical machining center for precision milling. Max spindle speed: 12000 RPM. Work envelope: 500 x 400 x 300 mm.

Safety

Always wear safety glasses. Lock out/tag out before maintenance. Never open the enclosure while spindle is running. Emergency stop (E-STOP) is located on the front panel.

Startup Procedure

1. Verify coolant level. 2. Power on main breaker. 3. Home all axes (G28). 4. Load program and verify tool offsets. 5. Run dry cycle at reduced feed.

Maintenance Schedule

Daily: clean chips, check coolant. Weekly: lubricate ways, inspect tool holder. Monthly: spindle runout check, backup parameters. Quarterly: ball screw inspection.

Troubleshooting

Alarm 101 (door open): close safety door. Alarm 205 (tool break): replace tool, recalibrate length offset. Poor surface finish: reduce feed, check tool wear.

Specifications

Power: 15 kW. Tool capacity: 24 stations. Control: Fanuc-compatible. Coolant: water-soluble, 6-8% concentration.

Emergency Contacts

- 1 Maintenance Lead: ext. 204
- 2 Plant Manager: ext. 101
- 3 Safety Officer: ext. 999